DO NOW — RECALL

Final mixed-paper readiness

Name: _____ Class: _____ Date: _____

1. Difference between MAR and MDR?

2. How to encode a negative integer in two's complement?

3. What does $O(n^2)$ describe?

4. Full adder A=1 B=1 carry-in=1. State sum and carry.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

MAR holds address; MDR transports data/instruction contents.

2. Model answer

Invert fixed-width positive representation then add one.

3. Model answer

Quadratic asymptotic growth.

4. Model answer

$1+1+1=3=11_2$; sum1 carry1.

By the end, you can…

- Apply Paper 1 and Paper 2 knowledge together
- Demonstrate precise independent working
- Identify the strongest remaining revision priority

Vocabulary — know these terms

readiness — Ability to apply without prompts.

synoptic — Knowledge across multiple topics.

method evidence — Reasoning supporting an answer.

NEW LEARNING

Read, annotate and explain

Identify

Retrieve knowledge across both components

Model

Show working in numeric and algorithm tasks

Apply

Check terminology and boundary cases

Evaluate

Prioritise a final independent improvement target

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Normalise positive $0.0101_2 \times 2^3$.

Explain the method

Shift mantissa one position left and reduce exponent by one.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Q1 [8]: In a simplified sequential fetch, PC=12 and memory address 12 stores instruction ADD 7. State MAR, MDR, CIR and next PC (assume no branch or interrupt). [4] Convert −5 to 8-bit two's complement. [4]

Question 2

Q2 [8]: Trace binary search for 11 in [1,3,5,7,9,11,13]. [4] Explain base case and recursion progress. [4]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Q3 [9]: Evaluate cloud storage versus USB-only storage for an inclusive school revision service; weigh access, reliability, security. [9]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Publishing every lesson guarantees each student is ready for assessment.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.